

Inverse Trigonometric Functions

Question1

If

$$\theta = \tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{7} \right) + \tan^{-1} \left(\frac{1}{13} \right) + \tan^{-1} \left(\frac{1}{21} \right) + \tan^{-1} \left(\frac{1}{31} \right)$$

, then $\tan \theta =$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

$$\frac{3}{5}$$

B.

$$1$$

C.

$$\frac{5}{7}$$

D.

$$\frac{7}{9}$$

Answer: C

Solution:

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{7} \right) + \tan^{-1} \left(\frac{1}{13} \right) + \tan^{-1} \left(\frac{1}{21} \right) + \tan^{-1} \left(\frac{1}{31} \right) \\ &= \sum_{k=1}^5 \tan^{-1} \left(\frac{1}{k^2 + k + 1} \right) \end{aligned}$$

$$\text{and } \tan^{-1}(k+1) - \tan^{-1}(k) = \tan^{-1}\left(\frac{1}{k^2 + k + 1}\right)$$

$$\begin{aligned} \therefore \theta &= \sum_{k=1}^5 (\tan^{-1}(k+1) - \tan^{-1}(k)) \\ &= \tan^{-1}(2) - \tan^{-1}(1) + \tan^{-1}(3) \\ &\quad - \tan^{-1}(2) + \tan^{-1}(4) - \tan^{-1}(3) \\ &\quad + \tan^{-1}(5) - \tan^{-1}(4) + \tan^{-1}(6) \\ &\quad - \tan^{-1}(5) \\ &= \tan^{-1}(6) - \tan^{-1}(1) \\ &= \tan^{-1}\left(\frac{6-1}{1+6 \times 1}\right) = \tan^{-1}\left(\frac{5}{7}\right) \\ \Rightarrow \tan \theta &= \frac{5}{7} \end{aligned}$$

Question2

If $\tan^{-1} x = \cot h^{-1} y = \log \sqrt{5}$, then $\tan^{-1}(xy) =$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

$$\frac{\pi}{4}$$

B.

$$\frac{\pi}{3}$$

C.

$$\frac{\pi}{6}$$

D.

$$\frac{3\pi}{4}$$

Answer: A

Solution:

$$\because \coth^{-1} y = \tanh^{-1} \left(\frac{1}{y} \right)$$

$$\text{so, } \tanh^{-1} x = \tanh^{-1} \left(\frac{1}{y} \right)$$

$$\Rightarrow x = \frac{1}{y}$$

$$\Rightarrow xy = 1$$

$$\text{Therefore, } \tan^{-1}(xy) = \tan^{-1}(1) = \frac{\pi}{4}$$

Question3

If $f(x) = 2 + |\sin^{-1} x|$ and $A = \{x \in R / f^1(x) \text{ exists} \}$, then $A =$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

$$\{0\}$$

B.

$$[-1, 1]$$

C.

$$(-\infty, -1) \cup (1, \infty)$$

D.

$$(-1, 0) \cup (0, 1)$$

Answer: D

Solution:

$$\because f(x) = 2 + |\sin^{-1} x|$$

Domain of $\sin^{-1} x$ is $x \in [-1, 1]$

So, the domain of $f(x)$ is $[-1, 1]$

and $|\sin^{-1} x|$ is not differentiable at $x = 0$

So, the set A where $f'(x)$ exists is

$$(-1, 1) - \{0\}$$

Hence, $f(x)$ is differentiable at all points in $(-1, 0) \cup (0, 1)$

Question4

$$\tan \left(2 \tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{7} \right) \right) =$$

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$\frac{1}{\sqrt{3}}$$

B.

$$\sqrt{3}$$

C.

$$1$$

D.

$$3/7$$

Answer: C

Solution:

We have,

$$\begin{aligned} & \tan \left(2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} \right) \\ &= \tan \left(\tan^{-1} \left(\frac{2/3}{1-1/9} \right) + \tan^{-1} \frac{1}{7} \right) \left\{ \because 2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2} \right\} \\ &= \tan \left[\tan^{-1} \left(\frac{3}{4} \right) + \tan^{-1} \left(\frac{1}{7} \right) \right] \\ &= \tan \left[\tan^{-1} \left(\frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \cdot \frac{1}{7}} \right) \right] \left\{ \because \tan^{-1} x + \tan^{-1} y \right\} \\ &\quad \left\{ = \tan^{-1} \left(\frac{x+y}{1-xy} \right) \right\} \\ &= \tan \left(\tan^{-1} \left(\frac{25}{25} \right) \right) = 1 \end{aligned}$$

Question5

$$\tanh^{-1} \left(\frac{1}{3} \right) + \coth^{-1}(3) =$$

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$\operatorname{sech}^{-1} \left(\frac{1}{3} \right)$$

B.

$$\operatorname{cosech}^{-1} \left(\frac{1}{3} \right)$$

C.

$$\cosh^{-1} \left(\frac{4}{3} \right)$$

D.

$$\sinh^{-1} \left(\frac{3}{4} \right)$$

Answer: D

Solution:

We have,

$$\begin{aligned} & \tanh^{-1}\left(\frac{1}{3}\right) + \coth^{-1}(3) \\ &= \tanh^{-1}\left(\frac{1}{3}\right) + \tanh^{-1}\left(\frac{1}{3}\right) \\ &= 2 \tanh^{-1}\left(\frac{1}{3}\right) \\ &= \sinh^{-1}\left(\frac{2 \times \frac{1}{3}}{1 - \frac{1}{9}}\right) = \sinh^{-1}\left(\frac{2/3}{8/9}\right) \\ & \left\{ \because 2 \tanh^{-1} x = \sinh^{-1}\left(\frac{2x}{1-x^2}\right) \right\} \\ &= \sinh^{-1}\left(\frac{3}{4}\right) \end{aligned}$$

Question6

If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $\left(\frac{d^2y}{dx^2}\right)_{x=2} = k$, then $25k =$

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$(-3)^2$$

B.

$$(-2)^3$$

C.

$$3$$

D.

$$(-2)^5$$

Answer: B

Solution:

Given, $y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$

$$\therefore \sin^{-1} \frac{2x}{1+x^2} = 2 \tan^{-1} x$$

$$\therefore y = 2 \tan^{-1} x \Rightarrow \frac{dy}{dx} = \frac{2}{1+x^2}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{(-2)2x}{(1+x^2)^2} = -\frac{4x}{(1+x^2)^2}$$

$$\Rightarrow \left. \frac{d^2y}{dx^2} \right|_{x=2} = -\frac{8}{25} = k$$

$$\therefore 25k = -8 = (-2)^3$$

Question7

If $f(x) = \sec^{-1} \left(\frac{1}{2x^2-1} \right)$ and $g(x) = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$, then the derivative of $f(x)$ with respect to $g(x)$ is

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$\frac{1+x^2}{4\sqrt{1-x^2}}$$

B.

$$\frac{(1-x^2)}{4\sqrt{1+x^2}}$$

C.

$$-\frac{4(1-x^2)}{\sqrt{1+x^2}}$$

D.

$$-\frac{4(1+x^2)}{\sqrt{1-x^2}}$$

Answer: D

Solution:

Given,

$$f(x) = \sec^{-1} \left(\frac{1}{2x^2 - 1} \right), g(x) = \tan^{-1} \left(\frac{\sqrt{1+x^2} - 1}{x} \right)$$

$$\text{put, } x = \cos \theta \quad x = \tan \phi$$

$$f(x) = \sec^{-1} \left(\frac{1}{\cos 2\theta} \right) \quad g(x) = \tan^{-1} \left(\frac{\sec \phi - 1}{\tan \phi} \right)$$

$$= \sec^{-1}(\sec 2\theta)$$

$$= 2\theta$$

$$= 2 \cos^{-1} x$$

$$\Rightarrow f'(x) = \frac{-2}{\sqrt{1-x^2}}$$

$$= \tan^{-1} \left(\frac{1 - \cos \phi}{\sin \phi} \right)$$

$$= \tan^{-1} \left(\frac{2 \sin^2 \phi/2}{2 \sin \phi/2 \cos \phi/2} \right)$$

$$= \tan^{-1}(\tan \phi/2)$$

$$= \phi/2$$

$$= \frac{1}{2} \tan^{-1} x$$

$$\Rightarrow g'(x) = \frac{1}{2(1+x^2)}$$

Differentiating $f(x)$ w.r.t $g(x)$ is

$$= \frac{f'(x)}{g'(x)} = \frac{-4(1+x^2)}{\sqrt{1-x^2}}$$

Question8

If $A = \left\{ x \in R / \sin^{-1} \left(\sqrt{x^2 + x + 1} \right) \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \right\}$ and $B = \left\{ y \in R / y = \sin^{-1} \left(\sqrt{x^2 + x + 1} \right), x \in A \right\}$, then

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

$$A \cap B \neq \phi$$

B.

$$A \cap B^C = [0, 1]$$

C.

$$A^C \cap B = \left[\frac{\pi}{3}, \frac{\pi}{2} \right]$$

D.

$$A \cup B = R - \left\{ [-1, 0] \cup \left[\frac{\pi}{3}, \frac{\pi}{2} \right] \right\}$$

Answer: C

Solution:

$$A = \left\{ x \in R / \sin^{-1} \left(\sqrt{x^2 + x + 1} \right) \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \right\}$$

and

$$B = \left\{ y \in R / y = \sin^{-1} \left(\sqrt{x^2 + x + 1} \right), x \in A \right\}$$

Since, domain of $\sin^{-1}(x)$ is $[-1, 1]$

So, domain of $\sin^{-1} \sqrt{x^2 + x + 1}$ is real, so

so

$$\begin{aligned} \sqrt{x^2 + x + 1} &\in [-1, 1] \\ \Rightarrow \sqrt{x^2 + x + 1} &\in [0, 1] \end{aligned}$$

($\because$ square root is always non-negative)

$$\begin{aligned} \Rightarrow x^2 + x + 1 &\leq 1 \\ \Rightarrow x^2 + x &\leq 0 \\ \Rightarrow x(x + 1) &\leq 0 \end{aligned}$$

So, $x \in [-1, 0]$ for set A

Thus, $A = [-1, 0]$

Now, for set B , since $x \in A = [-1, 0]$, the range values for $\sqrt{x^2 + x + 1}$ for $x \in [-1, 0]$ we will find.

$$\begin{aligned} \text{At } x = -1, \sqrt{(-1)^2 - 1 + 1} \\ = \sqrt{1 - 1 + 1} = 1 \end{aligned}$$

$$\text{At } x = 0, \sqrt{0^2 + 0 + 1} = \sqrt{0 + 1} = 1$$

$$\begin{aligned} \text{At } x = -\frac{1}{2}, \sqrt{\left(\frac{-1}{2}\right)^2 - \frac{1}{2} + 1} \\ = \sqrt{\frac{1}{4} - \frac{1}{2} + 1} \\ = \sqrt{\frac{3}{4}} \end{aligned}$$

$$\text{So, } \sqrt{x^2 + x + 1} \in \left[\sqrt{\frac{3}{4}}, 1 \right] = \left[\frac{\sqrt{3}}{2}, 1 \right]$$

$$\text{Thus, } y = \sin^{-1} \left(\sqrt{x^2 + x + 1} \right)$$

$$\in \left[\sin^{-1} \left(\frac{\sqrt{3}}{2} \right), \sin^{-1}(1) \right]$$

$$= \left[\frac{\pi}{3}, \frac{\pi}{2} \right]$$

$$\text{Hence, } B = \left[\frac{\pi}{3}, \frac{\pi}{2} \right]$$

$$\Rightarrow \text{Now, } A \cap B = [-1, 0] \cap \left[\frac{\pi}{3}, \frac{\pi}{2} \right] = \phi$$

So, $A \cap B \neq \phi$ is false.

$$\Rightarrow A \cap B^C = [-1, 0] \cap \left[\frac{\pi}{3}, \frac{\pi}{2} \right]^C \text{ and the intersection with any set outside this cannot be } [0, 1],$$

So, $A \cap B^C = [0, 1]$ is false.

$$\Rightarrow A^C \text{ is everything except } [-1, 0].$$

$$\text{Since } B = (\pi/3, \pi/2) \subset R/(-1, 0)$$

$$\Rightarrow A^C \cap B \text{ is true.}$$

$$\Rightarrow A \cup B = R - \{[-1, 0] \cup [\frac{\pi}{3}, \frac{\pi}{2}]\}$$

So, union of A and B is the complement of their own union.

That's impossible.

$$\text{Thus, } A^C \cap B = \left[\frac{\pi}{3}, \frac{\pi}{2} \right], \text{ option (c) is correct.}$$

Question9

The domain of the function,

$$f(x) = \sqrt{\log_e \left(\frac{1}{x^2 - 4x + 4} \right)} + \sin^{-1} (x^2 - 2) \text{ is}$$

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

$$[1, 3]$$

B.

$$[1, 3)$$

C.

$$[1, \sqrt{3}]$$

D.

$$[1, \sqrt{3})$$

Answer: C

Solution:

Given function,

$$f(x) = \sqrt{\log_e \left(\frac{1}{x^2 - 4x + 4} \right)} + \sin^{-1} (x^2 - 2)$$

Since, log is defined for positive values only.

$$\text{So, } \frac{1}{x^2 - 4x + 4} > 0$$

Also since, $x^2 - 4x + 4 = (x - 2)^2 \geq 0$ this holds for $x \neq 2$

$$\Rightarrow \ln \left(\frac{1}{(x-2)^2} \right) \geq 0$$

Now, we know that $\ln(t)$ is strictly increasing for $t > 0$ and $\ln(1) = 0$, the inequality

$\ln \frac{1}{(x-2)^2} \geq 0$ is equivalent to

$$\frac{1}{(x-2)^2} \geq 1 \Leftrightarrow (x-2)^2 \leq 1$$

$$\Rightarrow |x-2| \leq 1$$

$$\text{i.e., } 1 \leq x \leq 3$$

But log is undefined for $x = 2$

$$\therefore x \in [1, 3] - \{2\}$$

For $\sin^{-1} (x^2 - 2)$ to be real, $x^2 - 2 \in [-1, 1]$

$$\Rightarrow -1 \leq x^2 - 2 \leq 1$$

$$\Rightarrow 1 \leq x^2 \leq 3 \Rightarrow 1 \leq |x| \leq \sqrt{3}$$

$$\Rightarrow x \in [-\sqrt{3}, -1] \cup [1, \sqrt{3}] \Rightarrow x \in [1, \sqrt{3}]$$

Question10

If $\cot (\cos^{-1} x) = \sec \left\{ \tan^{-1} \left(\frac{a}{\sqrt{b^2-a^2}} \right) \right\}, b > a$ then $x =$

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

$$\frac{b}{\sqrt{2b^2-a^2}}$$

B.

$$\frac{a}{\sqrt{2b^2-a^2}}$$

C.

$$\frac{\sqrt{b^2-a^2}}{a}$$

D.

$$\frac{\sqrt{b^2-a^2}}{b}$$

Answer: A

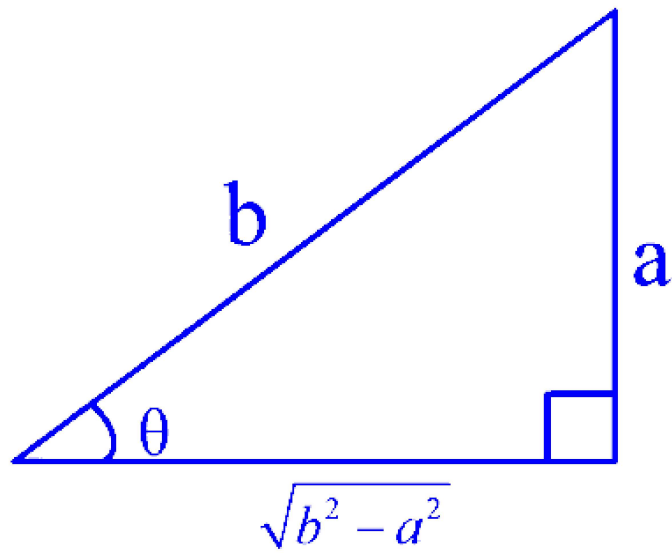
Solution:

Given,

$$\cot (\cos^{-1} x) = \sec \left\{ \tan^{-1} \left(\frac{a}{\sqrt{b^2-a^2}} \right) \right\}, b > a$$

$$\text{Let } \tan^{-1} \frac{a}{\sqrt{b^2-a^2}} = \theta \Rightarrow \tan \theta = \frac{a}{\sqrt{b^2-a^2}}$$

$$\sec \theta = \frac{b}{\sqrt{b^2-a^2}}$$



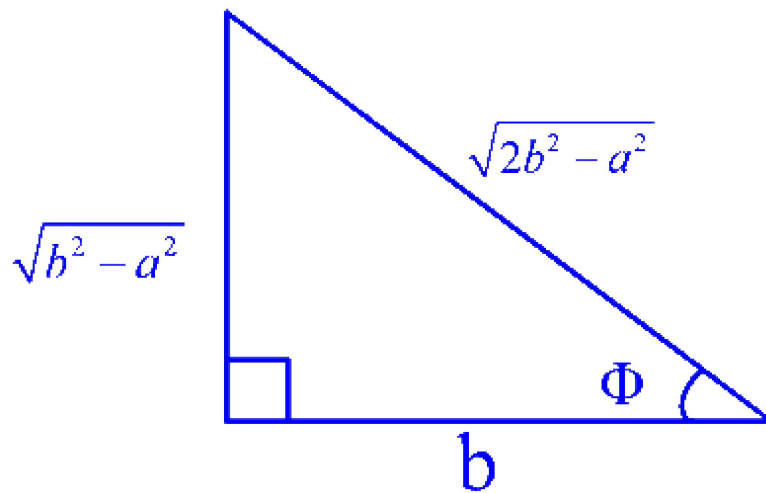
$$\therefore \cot(\cos^{-1} x) = \sec \left\{ \tan^{-1} \frac{a}{\sqrt{b^2 - a^2}} \right\}$$

$$\sec \theta = \frac{b}{\sqrt{b^2 - a^2}}$$

$$\Rightarrow \cos^{-1} x = \cot^{-1} \left(\frac{b}{\sqrt{b^2 - a^2}} \right)$$

$$\text{Again, let } \cot^{-1} \left(\frac{b}{\sqrt{b^2 - a^2}} \right) = \phi$$

$$\Rightarrow \cot \phi = \frac{b}{\sqrt{b^2 - a^2}}$$



$$\Rightarrow \cos \phi = \frac{b}{\sqrt{b^2 - a^2}}$$

$$\text{Now, } \cos^{-1} x = \phi$$

$$\Rightarrow x = \cos \phi = \frac{b}{\sqrt{2b^2 - a^2}}$$

$$\therefore x = \frac{b}{\sqrt{2b^2 - a^2}}$$

Question11

If $\sinh^{-1} x = \log 3$ and $\cosh^{-1} y = \log \frac{3}{2}$, then $\tanh^{-1}(x - y) =$

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

$$\log \sqrt{\frac{5}{3}}$$

B.

$$\log \frac{5}{3}$$

C.

$$\log \frac{4}{3}$$

D.

$$\log \frac{2}{\sqrt{3}}$$

Answer: A

Solution:

$$\text{Given, } \sinh^{-1} x = \log 3$$

$$\cosh^{-1} y = \log \frac{3}{2}$$

$$\therefore \sinh^{-1} x = \log(3)$$

$$\Rightarrow x = \sinh(\log 3)$$

$$= \frac{e^{\log 3} - e^{-\log 3}}{2}$$

$$= \frac{3 - \frac{1}{3}}{2} = \frac{8}{6} = \frac{4}{3}$$

$$\text{And } \cosh^{-1} y = \log \left(\frac{3}{2} \right)$$

$$\Rightarrow y = \cosh \left(\log \frac{3}{2} \right)$$

$$\Rightarrow \frac{e^{\log(\frac{3}{2})} + e^{-\log(\frac{3}{2})}}{2} = \frac{\frac{3}{2} + \frac{2}{3}}{2} = \frac{13}{12}$$

$$x - y = \frac{4}{3} - \frac{13}{12}$$

$$= \frac{16 - 13}{12} = \frac{3}{12} = \frac{1}{4}$$

Now, we know that

$$\tan h^{-1}(z) = \frac{1}{2} \ln \left(\frac{1+z}{1-z} \right)$$

Now, $z = x - y$

$$\text{So, } \tan h^{-1}(x - y) = \frac{1}{2} \ln \left(\frac{1+x-y}{1-(x-y)} \right)$$

$$= \frac{1}{2} \ln \left(\frac{1+x-y}{1-x+y} \right)$$

$$= \frac{1}{2} \ln \left(\frac{1+\frac{1}{4}}{1-\frac{1}{4}} \right) = \frac{1}{2} \ln \left(\frac{5/4}{3/4} \right)$$

$$= \frac{1}{2} \ln \left(\frac{5}{3} \right) = \ln \left(\sqrt{\frac{5}{3}} \right)$$

$$\text{So, } \tan h^{-1}(x - y) = \ln \left(\sqrt{\frac{5}{3}} \right)$$

Question12

The number of solution of

$$\tan^{-1} 1 + \frac{1}{2} \cos^{-1} x^2 - \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right) = 0 \text{ is}$$

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

3

B.

0

C.

1

D.

infinitely many

Answer: D

Solution:

$$\begin{aligned} & \tan^{-1}(1) + \frac{1}{2}\cos^{-1}(x^2) \\ & - \tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right) = 0 \\ & \because \tan^{-1}(1) = \pi/4 \text{ and let } \theta = \cos^{-1}(x^2) \\ & \Rightarrow x^2 = \cos \theta \\ & \text{and } 1+x^2 = 1+\cos \theta, 1-x^2 = 1-\cos \theta \\ & = \frac{\pi}{4} + \frac{\theta}{2} - \tan^{-1}\left(\frac{\sqrt{1+\cos \theta} + \sqrt{1-\cos \theta}}{\sqrt{1+\cos \theta} - \sqrt{1-\cos \theta}}\right) \\ & = \frac{\pi}{4} + \frac{\theta}{2} - \tan^{-1}\left(\frac{\sqrt{2}\cos \theta/2 + \sqrt{2}\sin \theta/2}{\sqrt{2}\cos \theta/2 - \sqrt{2}\sin \theta/2}\right) \\ & = \frac{\pi}{4} + \frac{\theta}{2} - \tan^{-1}\left(\frac{\cos \theta/2 + \sin \theta/2}{\cos \theta/2 - \sin \theta/2}\right) \\ & \because \frac{\cos x + \sin x}{\cos x - \sin x} = \tan\left(\frac{\pi}{4} + x\right) \\ & = \frac{\pi}{4} + \frac{\theta}{2} - \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right)\right) \\ & = \frac{\pi}{4} + \frac{\theta}{2} - \frac{\pi}{4} - \frac{\theta}{2} = 0 \end{aligned}$$

Hence, the identity is true for all x . Infinite many solutions.

Question 13

$$\tanh^{-1}(\sin \theta) =$$

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

$$\sinh^{-1}(\operatorname{cosec} \theta)$$

B.

$$\sinh^{-1}(\sec \theta)$$

C.

$$\cosh^{-1}(\operatorname{cosec} \theta)$$

D.

$$\cosh^{-1}(\sec \theta)$$

Answer: D

Solution:

$$\therefore \tanh^{-1}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \text{ (for } |x| < 1 \text{)}$$

$$\text{so, } \tanh^{-1}(\sin \theta) = \frac{1}{2} \ln \left(\frac{1+\sin \theta}{1-\sin \theta} \right)$$

$$\begin{aligned} \frac{1+\sin \theta}{1-\sin \theta} &= \frac{(1+\sin \theta)^2}{(1-\sin^2 \theta)} = \left(\frac{1+\sin \theta}{\cos \theta} \right)^2 \\ &= \frac{1}{2} \ln \left(\frac{1+\sin \theta}{\cos \theta} \right)^2 \\ &= \ln \left(\frac{1+\sin \theta}{\cos \theta} \right) \quad \dots (i) \end{aligned}$$

$$\text{and } \cosh^{-1}(x) = \ln \left(x + \sqrt{x^2 - 1} \right)$$

$$\begin{aligned} \therefore \cosh^{-1}(\sec \theta) &= \ln \left(\frac{1}{\cos \theta} + \sqrt{\frac{1-\cos^2 \theta}{\cos^2 \theta}} \right) \\ &= \ln \left(\frac{1+\sin \theta}{\cos \theta} \right) \quad \dots (ii) \end{aligned}$$

By comparing Eqs. (i) and (ii), we get

$$\tanh^{-1}(\sin \theta) = \cosh^{-1}(\sec \theta)$$

Question14

The interval in which the function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function is

Options:

A.

$$\left(0, \frac{\pi}{2}\right)$$

B.

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

C.

$$\left(-\frac{3\pi}{4}, \frac{\pi}{4}\right)$$

D.

$$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

Answer: C

Solution:

$$\because \sin x + \cos x = \sqrt{2} \sin \left(x + \frac{\pi}{4}\right)$$

$$\therefore f(x) = \tan^{-1} \left(\sqrt{2} \sin \left(x + \frac{\pi}{4}\right) \right)$$

$$\text{Now, } f'(x) = \frac{1}{1 + \left(\sqrt{2} \sin \left(x + \frac{\pi}{4}\right) \right)^2} \left(\sqrt{2} \cos \left(x + \frac{\pi}{4}\right) \right)$$

$$= \frac{\sqrt{2} \cos \left(x + \frac{\pi}{4}\right)}{1 + 2 \sin^2 \left(x + \frac{\pi}{4}\right)} \rightarrow \text{always} > 0$$

$$\text{for increasing } \sqrt{2} \cos \left(x + \frac{\pi}{4}\right) > 0$$

$$\Rightarrow \cos \left(x + \frac{\pi}{4}\right) > 0$$

$$\text{So, } -\frac{\pi}{2} + 2k\pi < x + \frac{\pi}{4} < \frac{\pi}{2} + 2k\pi$$

$$\Rightarrow -\frac{3\pi}{4} + 2k\pi < x < \frac{\pi}{4} + 2k\pi$$

for the principal interval,

$$\text{let } k = 0$$

Thus, the interval is

$$x \in \left(-\frac{3\pi}{4}, \frac{\pi}{4}\right)$$

Question15

The range of the real valued function $f(x) = \cos^{-1} \left(\frac{3}{\sqrt{9x^2 - 12x + 22}} \right)$ is

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

$$(0, \frac{\pi}{4}]$$

B.

$$[\frac{\pi}{4}, \frac{\pi}{2})$$

C.

$$[0, \pi]$$

D.

$$[\frac{\pi}{6}, \frac{\pi}{2})$$

Answer: B

Solution:

$$\text{Given, } f(x) = \cos^{-1} \left(\frac{3}{\sqrt{9x^2 - 12x + 22}} \right)$$

$$\text{Since, } 9x^2 - 12x + 22 = 9$$

$$\underbrace{\left[\left(x - \frac{2}{3} \right)^2 + 2 \right]}_{\geq 2} \geq 18$$

$$\Rightarrow \sqrt{9 \left(x - \frac{2}{3} \right)^2 + 18} \geq \sqrt{18}$$

$$\Rightarrow 0 < \frac{1}{\sqrt{9x^2 - 12x + 22}} \leq \frac{1}{\sqrt{18}}$$

$$\Rightarrow 0 < \frac{3}{\sqrt{9x^2 - 12x + 22}} \leq \frac{3}{\sqrt{18}}$$

$$\Rightarrow 0 < \frac{3}{\sqrt{9x^2 - 12x + 22}} \leq \frac{1}{\sqrt{2}}$$

$$\begin{aligned}\Rightarrow \cos^{-1} 0 &> \cos^{-1} \left(\frac{3}{\sqrt{9x^2 - 12x + 22}} \right) \\ &\geq \cos^{-1} \left(\frac{1}{\sqrt{2}} \right) \\ \Rightarrow \frac{\pi}{4} &\leq f(x) < \frac{\pi}{2}\end{aligned}$$

Question16

If the equation $2 \cot^{-1} (x^2 + 2x + k) = \pi - 3 \tan^{-1} (x^2 + 2x + k)$ has two distinct real solutions, then all the values of k lie in the interval

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

$$(-1, 2)$$

B.

$$(1, \infty)$$

C.

$$(-\infty, \infty)$$

D.

$$(-\infty, 1)$$

Answer: D

Solution:

As we know that,

$$\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$$

So, given equation reduces to

$$\begin{aligned}
& 2 \left(\frac{\pi}{2} - \tan^{-1} (x^2 + 2x + k) \right) \\
&= \pi - 3 \tan^{-1} (x^2 + 2x + k) \\
\Rightarrow & \tan^{-1} (x^2 + 2x + k) = 0 \\
\Rightarrow & x^2 + 2x + k = 0
\end{aligned}$$

Given Eq. (i) has two distinct real solution. So, its discriminant value is always greater than zero.

$$\text{So, } (4) -4k > 0$$

$$\Rightarrow k < 1 \Rightarrow k \in (-\infty, 1)$$

Question17

$$\sec h^{-1}(\sin \alpha) =$$

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

$$\log \left(\sin \alpha + \sqrt{\sin^2 \alpha - 1} \right)$$

B.

$$\log(\tan \alpha + 1)$$

C.

$$\log \left(\cot \frac{\alpha}{2} \right)$$

D.

$$\log \left(\frac{1+\tan \alpha}{2 \sin \alpha} \right)$$

Answer: C

Solution:

$$\begin{aligned}
\therefore \operatorname{sech}^{-1}(x) &= \ln \left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1} \right) \\
\therefore \operatorname{sech}^{-1}(\sin \alpha) &= \ln \left(\frac{1}{\sin \alpha} + \sqrt{\frac{1}{\sin^2 \alpha} - 1} \right) \\
&= \ln \left(\frac{1}{\sin \alpha} + \sqrt{\frac{1 - \sin^2 \alpha}{\sin^2 \alpha}} \right) \\
&= \ln \left(\frac{1}{\sin \alpha} + \frac{\cos \alpha}{\sin \alpha} \right) \\
&= \ln \left(\frac{1 + \cos \alpha}{\sin \alpha} \right) = \ln \left(\frac{2 \cos^2 \frac{\alpha}{2}}{2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}} \right) \\
&= \ln \left(\cot \frac{\alpha}{2} \right)
\end{aligned}$$

Question18

If $y = \log \left(\sec \left(\tan^{-1} x \right) \right) (x > 0)$, then $\frac{dy}{dx}$ at $x = 1$ is

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

1

B.

3

C.

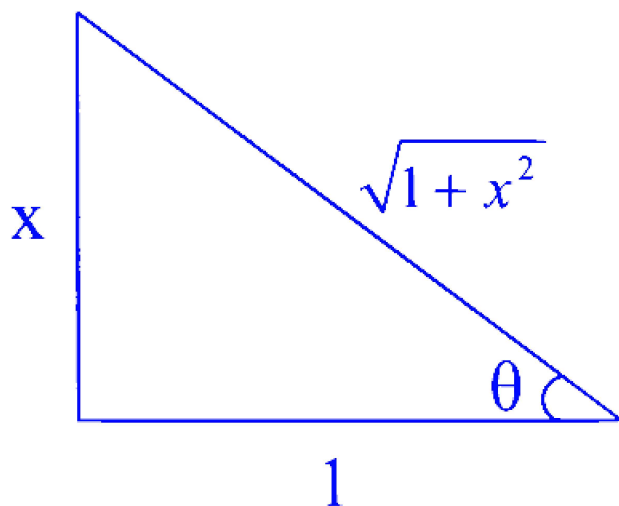
$\frac{1}{2}$

D.

$\frac{3}{2}$

Answer: C

Solution:



Given $y = \log (\sec (\tan ^{-1} x)), x > 0$

$$= \log \left(\sec \left(\sec ^{-1} \frac{\sqrt{1+x^2}}{1} \right) \right)$$

$$y = \log \left(\sqrt{1+x^2} \right)$$

$\therefore$ On differentiate w.r.t. x ,

$$\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}} \times \frac{1}{2\sqrt{1+x^2}} \times (2x) = \frac{x}{(1+x^2)^2}$$

$$\therefore \frac{dy}{dx} \Big|_{x=1} = \frac{1}{1+(1)^2} = \frac{1}{2}$$

Question19

If $y = \sin^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right)$ and $-\frac{3\pi}{2} < x < -\frac{\pi}{2}$, then $\frac{dy}{dx} =$

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

$$-\frac{\left| \operatorname{cosec} \frac{x}{2} \right|}{2\sqrt{\sin^2 \frac{x}{2} - \cos^2 \frac{x}{2}}}$$

B.

$$\frac{|\sec \frac{x}{2}|}{2\sqrt{\cos x}}$$

C.

$$\frac{\cos \frac{x}{2}}{2\sqrt{\cos x}}$$

D.

$$\frac{\cos \frac{x}{2}}{\sqrt{\cos x}}$$

Answer: B

Solution:

Given,

$$\begin{aligned} y &= \sin^{-1} \left(\frac{\sqrt{1 + \sin x} + \sqrt{1 - \sin x}}{\sqrt{1 + \sin x} - \sqrt{1 - \sin x}} \right) \\ \Rightarrow y &= \sin^{-1} \left(\frac{(\sqrt{1 + \sin x} + \sqrt{1 - \sin x})^2}{(\sqrt{1 + \sin x})^2 - (\sqrt{1 - \sin x})^2} \right) \\ &= \sin^{-1} \left(\frac{1 + \sin x + 1 - \sin x + 2\sqrt{1 - \sin^2 x}}{1 + \sin x - 1 + \sin x} \right) \\ &= \sin^{-1} \left(\frac{2 + 2|\cos x|}{2 \sin x} \right) = \sin^{-1} \left(\frac{1 - \cos x}{\sin x} \right) \left(\because \frac{-3\pi}{2} < x < \frac{-\pi}{2} \right) \\ &\quad \left(\because |\cos x| = -\cos x \right) \\ &= \sin^{-1} \left(\frac{2 \sin^2 \left(\frac{x}{2} \right)}{2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)} \right) = \sin^{-1} \left(\tan \frac{x}{2} \right) \\ \therefore \frac{dy}{dx} &= \frac{1}{\sqrt{1 - \tan^2 \left(\frac{x}{2} \right)}} \times \sec^2 \left(\frac{x}{2} \right) \times \frac{1}{2} \\ &= \frac{|\cos \frac{x}{2}|}{\sqrt{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}} \times \frac{1}{|\cos \frac{x}{2}|^2} \times \frac{1}{2} = \frac{|\sec \frac{x}{2}|}{2\sqrt{\cos x}} \end{aligned}$$

Question20

If $\frac{1}{2} \sin^{-1} \left(\frac{3 \sin 2\theta}{5 + 4 \cos 2\theta} \right) = \tan^{-1} x$, then $x =$

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

$$\tan \frac{\theta}{3}$$

B.

$$\frac{1}{3} \tan \theta$$

C.

$$\tan 3\theta$$

D.

$$\frac{1}{3} \tan 3\theta$$

Answer: B

Solution:

$$\begin{aligned} \frac{1}{2} \sin^{-1} \left(\frac{3 \sin 2\theta}{5 + 4 \cos 2\theta} \right) &= \tan^{-1} x \\ \Rightarrow \frac{1}{2} \sin^{-1} \left(\frac{3 \cdot \frac{2 \tan \theta}{1 + \tan^2 \theta}}{5 + 4 \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right)} \right) &= \tan^{-1} x \\ \Rightarrow \sin^{-1} \left(\frac{6 \tan \theta}{9 + \tan^2 \theta} \right) &= 2 \tan^{-1} x \\ \Rightarrow \sin^{-1} \left(\frac{2 \cdot \left(\frac{1}{3} \tan \theta \right)}{1 + \left(\frac{\tan \theta}{3} \right)^2} \right) &= \sin^{-1} \left(\frac{2x}{1 + x^2} \right) \\ \Rightarrow x &= \frac{1}{3} \tan \theta \end{aligned}$$

Question21

If $\operatorname{sech}^{-1} x = \log 2$ and $\operatorname{cosech}^{-1} y = -\log 3$, then $(x + y) =$

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

$$\frac{1}{6}$$

B.

$$\frac{1}{20}$$

C.

$$6$$

D.

$$20$$

Answer: B

Solution:

$$\operatorname{sech}^{-1} x = \log 2$$

$$\text{and } \operatorname{cosech}^{-1} y = -\log 3$$

$$\operatorname{sech}^{-1} x = \log 2$$

$$\Rightarrow \log \left(\frac{1 + \sqrt{1 - x^2}}{x} \right) = \log 2$$

$$\Rightarrow \frac{1 + \sqrt{1 - x^2}}{x} = 2$$

$$\Rightarrow 1 + \sqrt{1 - x^2} = 2x$$

$$\sqrt{1 - x^2} = 2x - 1$$

Squaring both sides, we get

$$\Rightarrow 1 - x^2 = 4x^2 + 1 - 4x$$

$$\Rightarrow 5x^2 - 4x = 0$$

$$\Rightarrow x(5x - 4) = 0$$

$$x = \frac{4}{5}$$

$$\therefore \operatorname{cosech}^{-1} y = -\log 3$$

$$\Rightarrow \log \left(\frac{1 + \sqrt{1 + y^2}}{y} \right) = \log \frac{1}{3}$$

$$\Rightarrow \frac{1 + \sqrt{1 + y^2}}{y} = \frac{1}{3}$$

$$\Rightarrow 1 + \sqrt{1 + y^2} = \frac{y}{3}$$

$$\Rightarrow \sqrt{1 + y^2} = \frac{y}{3} - 1$$

Squaring both sides, we get

$$\Rightarrow 1 + y^2 = \frac{y^2}{9} + 1 - \frac{2}{3}y$$

$$\Rightarrow \frac{8y^2}{9} + \frac{2}{3}y = 0$$

$$\Rightarrow \frac{2}{3}y \left(\frac{4y}{3} + 1 \right) = 0$$

$$\Rightarrow y = \frac{-3}{4}$$

$$\Rightarrow x + y = \frac{4}{5} - \frac{3}{4} = \frac{16 - 15}{20}$$

$$\Rightarrow x + y = \frac{1}{20}$$

Question22

If $y = \tan^{-1} \left(\frac{x}{1+2x^2} \right) + \tan^{-1} \left(\frac{x}{1+6x^2} \right)$, then $\frac{dy}{dx} =$

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

$$\frac{4}{16x^2+1} - \frac{3}{9x^2+1}$$

B.

$$\frac{3}{9x^2+1} - \frac{1}{x^2+1}$$

C.

$$\frac{3}{9x^2+1} - \frac{2}{4x^2+1}$$

D.

$$\frac{1}{9x^2+1} - \frac{1}{x^2+1}$$

Answer: B

Solution:

$$\begin{aligned}
 y &= \tan^{-1} \left(\frac{x}{1+2x^2} \right) + \tan^{-1} \left(\frac{x}{1+6x^2} \right) \\
 \Rightarrow y &= \tan^{-1} 2x - \tan^{-1} x + \tan^{-1} 3x \\
 \Rightarrow y &= \tan^{-1} 3x - \tan^{-1} x \\
 \Rightarrow \frac{dy}{dx} &= \frac{1 \times 3}{1+9x^2} - \frac{1}{1+x^2} \\
 \Rightarrow \frac{dy}{dx} &= \frac{3}{9x^2+1} - \frac{1}{x^2+1}
 \end{aligned}$$

Question23

The range of the real valued function

$$f(x) = \cos^{-1}(-x) + \sin^{-1}(-x) + \operatorname{cosec}^{-1}(x) \text{ is}$$

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

$$\left\{0, \frac{\pi}{2}\right\}$$

B.

$$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$$

C.

$$\left(0, \frac{\pi}{2}\right)$$

D.

$$\{0, \pi\}$$

Answer: B

Solution:

$$f(x) = \cos^{-1}(-x) + \sin^{-1}(-x) + \operatorname{cosec}^{-1}(x)]$$

$$\text{Domain} = (-1, 1)$$

$$\Rightarrow f(x) = \pi - \cos^{-1} x - \sin^{-1} x + \operatorname{cosec}^{-1} x$$

$$\Rightarrow f(x) = \pi - \frac{\pi}{2} + \operatorname{cosec}^{-1} x$$

$$\Rightarrow f(x) = \frac{\pi}{2} + \operatorname{cosec}^{-1} x$$

Domain of $\operatorname{cosec}^{-1} x$

$$(-\infty, -1] \cup [1, \infty)$$

The range of $\operatorname{cosec}^{-1} x$ is

$$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$$

$$\therefore f(x) = \frac{\pi}{2} + \operatorname{cosec}^{-1} x$$

$\therefore$ Range of

$$\begin{aligned} f(x) &= \left[\frac{\pi}{2} - \frac{\pi}{2}, \frac{\pi}{2} + 0\right) \cup \left(0 + \frac{\pi}{2}, \frac{\pi}{2} + \frac{\pi}{2}\right] \\ &= \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right] \end{aligned}$$

Question24

The horizontal distance between a tower and a building is $10\sqrt{3}$ units. If the angle of depression of the foot of the building from the top of the tower is 60° and the angle of elevation of the top of the building from the foot of the tower is 30° , then the sum of the heights of the tower and the building is

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

60

B.

50

C.

40

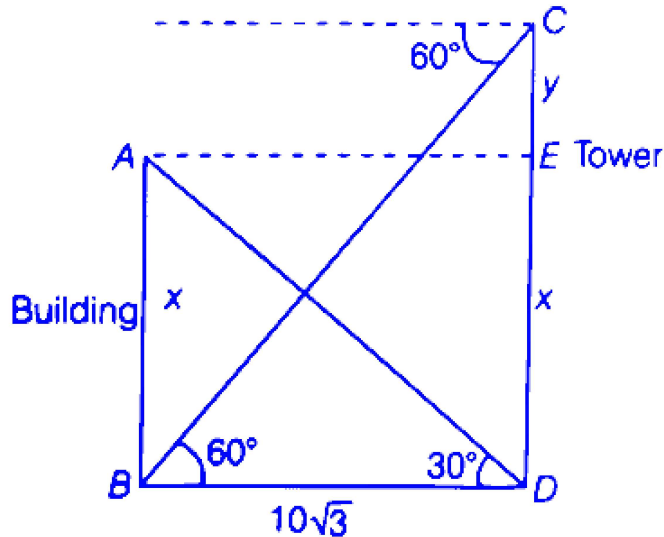
D.

30

Answer: C

Solution:

In $\triangle ADB$,



$$\Rightarrow \tan 30^\circ = \frac{x}{10\sqrt{3}}$$

$$x = 10$$

... (i)

$$\triangle CDB, \tan 60^\circ = \frac{x+y}{10\sqrt{3}}$$

$$x+y = 30$$

$$\therefore \text{Sum of height} = 10 + 30 = 40$$

Question25

If x is a real number, then the number of solutions of $\tan^{-1}(\sqrt{x(x+1)}) + \sin^{-1}(\sqrt{x^2+x+1}) = \frac{\pi}{2}$ is

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

1

B.

2

C.

3

D.

4

Answer: B

Solution:

$$\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \left(\sqrt{x^2 + x + 1} \right) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1} \sqrt{x(x+1)} = \cos^{-1} \left(\sqrt{x^2 + x + 1} \right)$$

$$\Rightarrow \tan^{-1} \sqrt{x(x+1)} = \tan^{-1} \sqrt{\frac{-x^2 - x}{x^2 + x + 1}}$$

$$\Rightarrow \sqrt{x(x+1)} = \sqrt{\frac{-x^2 - x}{x^2 + x + 1}}$$

$$\Rightarrow x(x+1) = \frac{-x(x+1)}{(x^2 + x + 1)}$$

So, $x(x+1) = 0$ or $x^2 + x + 2 = 0$ (rejected)

$$\Rightarrow x(x+1) = 0 \Rightarrow x = 0, -1$$

$\therefore$ Only two real solution.

Question26

If $y = \tanh^{-1} \sqrt{\frac{1-x}{1+x}}$, then $\frac{dy}{dx} =$

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

$$-\frac{1}{2\sqrt{1-x^2}}$$

B.

$$\frac{-1}{2x\sqrt{1-x^2}}$$

C.

$$\frac{2}{1+x^2}$$

D.

$$\frac{1}{2x\sqrt{1+x^2}}$$

Answer: B

Solution:

$$\begin{aligned} y &= \tan h^{-1} \sqrt{\frac{1-x}{1+x}} \\ \therefore \frac{d}{dx} \tan h^{-1} x &= \frac{1}{1-x^2} \\ \therefore \frac{dy}{dx} &= \frac{1}{1-\frac{1-x}{1+x}} \times \frac{1}{2} \sqrt{\frac{1+x}{1-x}} \times \frac{(1+x)(-1) - (1-x)}{(1+x)^2} \\ &= \frac{1+x}{1+x-1+x} \times \frac{1}{2} \frac{\sqrt{1+x}}{\sqrt{1-x}} \times \frac{-2}{(1+x)^2} \\ &= -\frac{1+x}{2x} \times \frac{1}{(1+x)\sqrt{(1+x)(1-x)}} \\ &= -\frac{1}{2x\sqrt{1-x^2}} \end{aligned}$$

Question27

$$\tan^{-1} \frac{\sqrt{8-2\sqrt{15}}}{\sqrt{15}+1} + \tan^{-1} \frac{1}{\sqrt{5}} =$$

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

$$\frac{\pi}{6}$$

B.

$$\frac{\pi}{4}$$

C.

$$\frac{\pi}{3}$$

D.

$$\frac{\pi}{2}$$

Answer: A

Solution:

$$\begin{aligned}
 & \tan^{-1} \frac{\sqrt{8-2\sqrt{15}}}{\sqrt{15}+1} + \tan^{-1} \frac{1}{\sqrt{5}} \\
 &= \tan^{-1} \left(\frac{\sqrt{5}-\sqrt{3}}{\sqrt{15}+1} \right) + \tan^{-1} \left(\frac{1}{\sqrt{5}} \right) \\
 &= \tan^{-1} \left(\frac{\sqrt{5}-\sqrt{3}}{\sqrt{15}+1} \times \frac{\sqrt{15}-1}{\sqrt{15}-1} \right) + \tan^{-1} \frac{1}{\sqrt{5}} \\
 &= \tan^{-1} \left(\frac{5\sqrt{3}-3\sqrt{5}-\sqrt{5}+\sqrt{3}}{14} \right) + \tan^{-1} \frac{1}{\sqrt{5}} \\
 &= \tan^{-1} \left(\frac{6\sqrt{3}-4\sqrt{5}}{14} \right) + \tan^{-1} \frac{1}{\sqrt{5}} \\
 &= \tan^{-1} \left(\frac{3\sqrt{3}-2\sqrt{5}}{7} \right) + \tan^{-1} \frac{1}{\sqrt{5}} \\
 &= \tan^{-1} \left(\frac{\frac{3\sqrt{3}-2\sqrt{5}}{7} + \frac{1}{\sqrt{5}}}{1 - \frac{3\sqrt{3}-2\sqrt{5}}{7} \times \frac{1}{\sqrt{5}}} \right) \\
 &= \tan^{-1} \left(\frac{3\sqrt{15}-10+7}{7\sqrt{5}-3\sqrt{3}+2\sqrt{5}} \right) \\
 &= \tan^{-1} \left(\frac{3\sqrt{15}-3}{9\sqrt{5}-3\sqrt{3}} \right) \\
 &= \tan^{-1} \left(\frac{\sqrt{15}-1}{3\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \right) \\
 &= \tan^{-1} \left(\frac{3\sqrt{5}-\sqrt{3}}{3\sqrt{5}-\sqrt{3}} \times \frac{1}{\sqrt{3}} \right) \\
 &= \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}
 \end{aligned}$$

Question28

The derivative of $\sec^{-1} \left(\frac{1}{2x^2-1} \right)$ with respect to $\sqrt{1-x^2}$ at $x = \frac{1}{2}$ is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

-2

B.

1

C.

2

D.

4

Answer: D

Solution:

$$\sec^{-1} \frac{1}{2x^2-1} = \cos^{-1} (2x^2-1) = 2 \cos^{-1} x$$

$$\therefore f(x) =$$

$$\Rightarrow f'(x) = -\frac{2 \cos^{-1} x}{\sqrt{1-x^2}}$$

$$\Rightarrow g(x) = \sqrt{1-x^2}$$

$$\Rightarrow g'(x) = \frac{1}{2\sqrt{1-x^2}} \times (-2x)$$

$$= -\frac{x}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{df}{dg} = \frac{-\frac{2}{\sqrt{1-x^2}}}{-\frac{x}{\sqrt{1-x^2}}} = \frac{2}{x}$$

$$\Rightarrow \left(\frac{df}{dg} \right)_{\text{at } x=\frac{1}{2}} = \frac{2}{\frac{1}{2}} = 4$$

Question29

If $0 < x < \frac{1}{2}$ and $\alpha = \sin^{-1} x + \cos^{-1} \left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2} \right)$, then $\tan \alpha + \cot \alpha$ is equal to

AP EAPCET 2024 - 23th May Morning Shift

Options:

A. $\frac{4}{\sqrt{3}}$

B. $4\sqrt{3}$

C. $\frac{4x}{1-x^2}$

D. $x\sqrt{1-x^2}$

Answer: A

Solution:

We have, \$0

$$\begin{aligned}\alpha &= \sin^{-1} x + \cos^{-1} \left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2} \right) \\ &= \sin^{-1} x + \cos^{-1} \left(\frac{x}{2} + \frac{\sqrt{3}}{2} \sqrt{1-x^2} \right)\end{aligned}$$

Let $x = \sin \theta$

$$\begin{aligned}\alpha &= \sin^{-1}(\sin \theta) + \cos^{-1} \\ &= \theta + \cos^{-1} \left(\cos \left(\frac{\pi}{6} - \theta \right) \right) \\ &= \theta + \frac{\pi}{6} - \theta = \frac{\pi}{6} \\ \tan \alpha + \cot \alpha &\Rightarrow \tan \frac{\pi}{6} + \cot \frac{\pi}{6} \\ &= \frac{1}{\sqrt{3}} + \sqrt{3} = \frac{4}{\sqrt{3}}\end{aligned}$$

Question30

$\cot \left(\sum_{n=1}^{50} \tan^{-1} \left(\frac{1}{1+n+n^2} \right) \right)$ is equal to

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. $\frac{26}{25}$

B. $\frac{25}{26}$

C. $\frac{50}{51}$

D. $\frac{52}{51}$

Answer: A

Solution:

We have,

$$\cot \left[\sum_{n=1}^{50} \tan^{-1} \left(\frac{1}{1+n+n^2} \right) \right] \quad \dots (i)$$

$$\sum_{n=1}^{50} \tan^{-1} \left(\frac{1}{1+n+n^2} \right)$$

$$= \sum_{n=1}^{50} \tan^{-1} \left(\frac{(n+1) - n}{1+n(n+1)} \right)$$

$$= \sum_{n=1}^{50} [\tan^{-1}(n+1) - \tan^{-1} n]$$

$$= \tan^{-1} 51 - \tan^{-1} 1$$

$$= \tan^{-1} \left(\frac{51-1}{1+51 \cdot 1} \right) = \tan^{-1} \left(\frac{50}{52} \right)$$

Now, from Eq. (i), we get

$$\begin{aligned} \Rightarrow \cot \left(\tan^{-1} \left(\frac{50}{52} \right) \right) &= \cot \left(\cot^{-1} \left(\frac{52}{50} \right) \right) \\ &= \frac{52}{50} = \frac{26}{25} \end{aligned}$$

Question31

The value of x such that $\sin\left(2 \tan^{-1} \frac{3}{4}\right) = \cos\left(2 \tan^{-1} x\right)$

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. 7

B. $\frac{3}{7}$

C. $\frac{1}{7}$

D. $\frac{4}{7}$

Answer: C

Solution:

To find the value of x , we need to equate:

$$\sin\left(2 \tan^{-1} \frac{3}{4}\right) = \cos\left(2 \tan^{-1} x\right).$$

Step-by-step Solution:

1. Left-hand side (LHS):

Let $\theta = \tan^{-1} \frac{3}{4}$, which implies $\tan \theta = \frac{3}{4}$.

We have:

$$\sin(2\theta) = \frac{2 \tan \theta}{1 + \tan^2 \theta} = \frac{2 \times \frac{3}{4}}{1 + \left(\frac{3}{4}\right)^2}$$

Calculating further:

$$= \frac{\frac{3}{2}}{1 + \frac{9}{16}} = \frac{\frac{3}{2}}{\frac{25}{16}} = \frac{24}{25}.$$

2. Right-hand side (RHS):

Let $z = \tan^{-1} x$, hence $x = \tan z$.

Therefore:

$$\cos(2z) = \frac{1 - \tan^2 z}{1 + \tan^2 z} = \frac{1 - x^2}{1 + x^2}.$$

3. Equating LHS to RHS:

$$\frac{24}{25} = \frac{1 - x^2}{1 + x^2}.$$

Cross-multiplying gives:

$$24 + 24x^2 = 25 - 25x^2.$$

Simplifying:

$$49x^2 = 1,$$

which leads to:

$$x^2 = \frac{1}{49} = \frac{1}{7^2},$$

thus:

$$x = \frac{1}{7}.$$

Question32

The range of the real valued function

$$f(x) = \sin^{-1} \left(\frac{1+x^2}{2x} \right) + \cos^{-1} \left(\frac{2x}{1+x^2} \right) \text{ is}$$

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. $\left\{ \frac{\pi}{2} \right\}$

B. R

C. Q

D. $\left\{ -\frac{\pi}{2}, \frac{\pi}{2} \right\}$

Answer: A

Solution:

Given that

$$f(x) = \sin^{-1} \left(\frac{1+x^2}{2x} \right) + \cos^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$\text{Let } t = \frac{1+x^2}{2x} \text{ and } u = \frac{2x}{1+x^2}$$

$$\therefore t = \frac{1}{u}$$

$$\therefore t \cdot u \equiv 1$$

We know that

$$\sin^{-1}(a) + \cos^{-1}(a) = \frac{\pi}{2}, a \in [-1, 1]$$

$$\text{Thus, } \sin^{-1}(t) + \cos^{-1}(u) = \frac{\pi}{2}$$

So, the range of the function

$$f(x) = \sin^{-1} \left(\frac{1+x^2}{2x} \right) + \cos^{-1} \left(\frac{2x}{1+x^2} \right) \text{ is } \left\{ \frac{\pi}{2} \right\}$$

Question33

The real values of x that satisfy the equation $\tan^{-1} x + \tan^{-1} 2x = \frac{\pi}{4}$ is

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. $\frac{-3 \pm \sqrt{17}}{4}$

B. $-1 \pm \sqrt{3}$

C. $\sqrt{3} - 1$

D. $\frac{\sqrt{17}-3}{4}$

Answer: D

Solution:

Given $\tan^{-1} x + \tan^{-1} 2x = \frac{\pi}{4}$

We know that,

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right) \text{ where}$$

$$xy < 1$$

Now, $\tan^{-1} x + \tan^{-1} 2x = \frac{\pi}{4}$

$$\Rightarrow \tan^{-1} \left(\frac{x+2x}{1-x \cdot 2x} \right) = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1} \left(\frac{3x}{1-2x^2} \right) = \frac{\pi}{4}$$

$$\Rightarrow \frac{3x}{1-2x^2} = \tan \frac{\pi}{4}$$

$$\Rightarrow \frac{3x}{1-2x^2} = 1$$

$$\Rightarrow \frac{3x}{1-2x^2} = 1 \quad \left[\because \tan \frac{\pi}{4} = 1 \right]$$

$$\Rightarrow 3x = 1 - 2x^2$$

$$\Rightarrow 2x^2 + 3x - 1 = 0$$

Using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Where, $a = 2, b = 3, c = -1$

$$x = \frac{-3 \pm \sqrt{9 - 4 \times 2 \times (-1)}}{4}$$

$$= \frac{-3 \pm \sqrt{9 + 8}}{4}$$

$$\Rightarrow x = \frac{-3 \pm \sqrt{17}}{4}$$

Thus, $x = \frac{-3 + \sqrt{17}}{4}$, or $\frac{-3 - \sqrt{17}}{4}$

$$\text{For, } x = \frac{-3 - \sqrt{17}}{4}$$

Since, $x \cdot 2x < 1 \Rightarrow 2x^2 < 1$

$$\Rightarrow 2\left(\frac{-3 - \sqrt{17}}{4}\right)^2 < 1, \text{ which is not true.}$$

$\therefore x = \frac{-3 - \sqrt{17}}{4}$ is not possible

$$\text{For } x = \left(\frac{-3 + \sqrt{17}}{4}\right)$$

Now, $2x^2 < 1$

$$2\left(\frac{-3 + \sqrt{17}}{4}\right)^2 < 1$$

Hence, $x = \frac{\sqrt{17} - 3}{4}$ is the only solution of the given equation.

Question34

$$2 \coth^{-1}(4) + \sec h^{-1}\left(\frac{3}{5}\right) =$$

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. $\log 5$

B. $2 \log 3$

C. $3 \log 2$

D. $\log \frac{5}{3}$

Answer: A

Solution:

We have, $2 \coth^{-1}(4) + \operatorname{sech}^{-1}\left(\frac{3}{5}\right)$

We know that $\coth^{-1}(x) = \frac{1}{2} \log \left(\frac{x+1}{x-1} \right)$

$$\operatorname{sech}^{-1}(x) = \log \left(\frac{1 + \sqrt{1 - x^2}}{x} \right), 0 < x \leq 1$$

$$\begin{aligned} \therefore 2 \coth^{-1}(4) + \operatorname{sech}^{-1}\left(\frac{3}{5}\right) &= 2 \times \frac{1}{2} \log \left(\frac{4+1}{4-1} \right) + \log \left(\frac{1 + \sqrt{1 - \left(\frac{3}{5}\right)^2}}{\frac{3}{5}} \right) \\ &= \log \left(\frac{5}{3} \right) + \log \left(\frac{1 + \frac{1}{5} \sqrt{25 - 9}}{\frac{3}{5}} \right) \\ &= [\log 5 - \log 3] + \log \left(\frac{5 + \sqrt{16}}{3} \right) \\ &= [\log 5 - \log 3] + \log \left(\frac{5 + 4}{3} \right) \\ &= [\log 5 - \log 3] + \log \left(\frac{9}{3} \right) \\ &= [\log 5 - \log 3] + \log 3 \\ &= \log 5 \end{aligned}$$

Question35

If $y = \sin^{-1} x$, then $(1 - x^2)y_2 - xy_1 =$

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. 0

B. 1

C. 2

D. $2y$

Answer: A

Solution:

If $y = \sin^{-1} x$, then

$$(1 - x^2)y_2 - xy_1 = ?$$

We have, $\Rightarrow y = \sin^{-1} x$

On differentiating y w.r.t. x , we get

$$\begin{aligned}\Rightarrow \frac{dy}{dx} &= y_1 = \frac{d}{dx} (\sin^{-1}(x)) \\ \Rightarrow \frac{dy}{dx} &= y_1 = \frac{1}{\sqrt{1-x^2}} \quad \dots (i)\end{aligned}$$

Again differentiate Eq. (i) w.r.t. x , we get

$$\begin{aligned}\Rightarrow y_2 &= \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{1}{\sqrt{1-x^2}} \right) \\ \Rightarrow y_2 &= \frac{x}{\sqrt{1-x^2} (1-x^2)} \\ \Rightarrow y_2 (1-x^2) &= \frac{x}{\sqrt{1-x^2}}\end{aligned}$$

From Eq. (i), we get

$$\begin{aligned}y_2 (1-x^2) &= xy_1 \\ \Rightarrow y_2 (1-x^2) - xy_1 &= 0\end{aligned}$$

Question36

If $\cos^{-1} 2x + \cos^{-1} 3x = \frac{\pi}{3}$ and $4x^2 = \frac{a}{b}$, then $a + b$ is equal to

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. 12

B. 11

C. 31

D. 10

Answer: D

Solution:

$$\text{Given, } \cos^{-1} 2x + \cos^{-1} 3x = \frac{\pi}{3}$$

$$\begin{aligned}\cos^{-1} \left[(2x)(3x) - \sqrt{1 - (2x)^2} \sqrt{1 - (3x)^2} \right] &= \frac{\pi}{3} \\ \Rightarrow 6x^2 - \sqrt{1 - 4x^2} \sqrt{1 - 9x^2} &= \cos \frac{\pi}{3} \\ \Rightarrow 6x^2 - \frac{1}{2} &= \sqrt{(1 - 4x^2)(1 - 9x^2)}\end{aligned}$$

On squaring both sides, we get

$$\begin{aligned}36x^4 + \frac{1}{4} - 2 \cdot (6x^2) \frac{1}{2} &= 1 - 9x^2 - 4x^2 + 36x^4 \\ \Rightarrow \frac{1}{4} - 6x^2 &= 1 - 13x^2 \Rightarrow 7x^2 = \frac{3}{4} \\ \Rightarrow x^2 &= \frac{3}{28} \Rightarrow x = \frac{\sqrt{3}}{2\sqrt{7}} \\ 4x^2 &= 4 \times \frac{3}{28} = \frac{3}{7} = \frac{a}{b} \quad [\text{given}]\end{aligned}$$

$$\text{Thus, } a + b = 3 + 7 = 10$$

Question 37

If $\theta = \sec^{-1}(\cosh u)$, then $u =$

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. $\log_e \left(\cot \left(\frac{\theta}{2} - \frac{\pi}{4} \right) \right)$

B. $\log_e \left(\tan \left(\frac{\theta}{2} - \frac{\pi}{4} \right) \right)$

C. $\log_\theta \left(\tan \left(\frac{\pi}{4} - \frac{\theta}{2} \right) \right)$

D. $\log_e \left(\tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right)$

Answer: D

Solution:

$$\text{Given, } \theta = \sec^{-1}(\cosh u)$$

$$\Rightarrow \sec \theta = \cosh u$$

$$\Rightarrow u = \cosh^{-1}(\sec \theta)$$

$$\text{Since, } \cosh^{-1}(x) = \log \left(x + \sqrt{x^2 - 1} \right)$$

$$\cosh^{-1}(\sec \theta) = \log \left(\sec \theta + \sqrt{\sec^2 \theta - 1} \right)$$

$$= \log(\sec \theta + \tan \theta)$$

$$= \log \left(\frac{1 + \sin \theta}{\cos \theta} \right)$$

$$= \log \left(\frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}} \right)$$

$$= \log \left(\frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} \right)$$

$$= \log \left(\frac{\tan \frac{\pi}{4} + \tan \frac{\theta}{2}}{1 - \tan \frac{\pi}{4} \tan \frac{\theta}{2}} \right)$$

$$= \log \left(\tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right)$$

$$\therefore u = \log \left(\tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right).$$

Question38

If $\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8} = k$, then

$$\sin^{-1} \left(\sqrt{\frac{k}{2}} \right) + \cos^{-1} \left(\frac{k}{3} \right) =$$

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $\frac{2\pi}{3}$

B. $\frac{3\pi}{4}$

C. $\frac{\pi}{4}$

D. $\frac{\pi}{2}$

Answer: A

Solution:

We are given:

$$\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8} = k$$

This can be rewritten as:

$$\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + [\cos(\pi - \frac{3\pi}{8})]^4 + [\cos(\pi - \frac{\pi}{8})]^4 = k$$

Since $\cos(\pi - \theta) = -\cos(\theta)$, the expression simplifies to:

$$\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{\pi}{8} = k$$

Thus, we have:

$$2 \cos^4 \frac{\pi}{8} + 2 \cos^4 \frac{3\pi}{8} = k$$

Next, using the values:

$$\cos \frac{\pi}{8} = \sqrt{\frac{2+\sqrt{2}}{4}} \quad \text{and} \quad \cos \frac{3\pi}{8} = \sqrt{\frac{2-\sqrt{2}}{4}}$$

Thus:

$$2 \left[\left(\frac{2+\sqrt{2}}{4} \right)^2 + \left(\frac{2-\sqrt{2}}{4} \right)^2 \right] = k$$

Expanding these squares:

$$\left(\frac{2+\sqrt{2}}{4} \right)^2 = \frac{4+2\sqrt{2}+1}{16} = \frac{6+2\sqrt{2}}{16}$$

$$\left(\frac{2-\sqrt{2}}{4} \right)^2 = \frac{4-2\sqrt{2}+1}{16} = \frac{6-2\sqrt{2}}{16}$$

Adding these together:

$$\frac{6+2\sqrt{2}}{16} + \frac{6-2\sqrt{2}}{16} = \frac{12}{16} = \frac{3}{4}$$

Thus:

$$k = 2 \times \frac{3}{4} = \frac{3}{2}$$

Now, we compute:

$$\sin^{-1} \left(\sqrt{\frac{k}{2}} \right) + \cos^{-1} \left(\frac{k}{3} \right)$$

$$= \sin^{-1} \left(\sqrt{\frac{3}{4}} \right) + \cos^{-1} \left(\frac{1}{2} \right)$$

$$= \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) + \cos^{-1} \left(\frac{1}{2} \right)$$

$$= \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$$

Question39

$$4 \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{1}{70} + \tan^{-1} \frac{1}{99} =$$

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $\frac{\pi}{12}$

B. $\frac{\pi}{6}$

C. $\frac{\pi}{4}$

D. $\frac{\pi}{3}$

Answer: C

Solution:

We are given the expression:

$$4 \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{1}{70} + \tan^{-1} \frac{1}{99}$$

Let's simplify this expression step-by-step:

First, break the expression:

$$4 \tan^{-1} \frac{1}{5} = 2 \left(\tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{5} \right)$$

Use the addition formula for arctan:

$$\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$$

Applying this to $\tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{5}$:

$$= \tan^{-1} \left(\frac{\frac{1}{5} + \frac{1}{5}}{1 - \frac{1}{5} \cdot \frac{1}{5}} \right) = \tan^{-1} \left(\frac{\frac{2}{5}}{\frac{24}{25}} \right) = \tan^{-1} \left(\frac{5}{12} \right)$$

$$\text{Therefore, } 2 \left(\tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{5} \right) = 2 \tan^{-1} \left(\frac{5}{12} \right)$$

Now incorporate the other terms:

$$2 \tan^{-1} \left(\frac{5}{12} \right) + \tan^{-1} \left(\frac{1}{99} \right) - \tan^{-1} \left(\frac{1}{70} \right)$$

Simplify further using the arctan subtraction formula:

$$\tan^{-1} a - \tan^{-1} b = \tan^{-1} \left(\frac{a-b}{1+ab} \right)$$

$$\tan^{-1} \left(\frac{1}{99} \right) - \tan^{-1} \left(\frac{1}{70} \right) = \tan^{-1} \left(\frac{\frac{1}{99} - \frac{1}{70}}{1 + \frac{1}{99} \cdot \frac{1}{70}} \right)$$

$$= \tan^{-1} \left(\frac{-29}{6931} \right)$$

Final simplification:

$$2 \tan^{-1} \left(\frac{5}{12} \right) - \tan^{-1} \left(\frac{1}{239} \right)$$

$$\tan^{-1} \left(\frac{\frac{5}{6}}{1 - \frac{25}{144}} \right) - \tan^{-1} \left(\frac{29}{6931} \right)$$

Combine using $\tan^{-1} \left(\frac{a-b}{1+ab} \right)$:

$$\tan^{-1} \left(\frac{120}{119} \right) - \tan^{-1} \left(\frac{1}{239} \right)$$

Arrive at the final form:

Using the formula:

$$\tan^{-1} \left(\frac{120 \times 239 - 119}{119 \times 239 + 120} \right) = \tan^{-1}(1) = \frac{\pi}{4}$$

Thus, the simplified result of the given expression is $\frac{\pi}{4}$.

Question40

$$\cosh \left(\sinh^{-1}(\sqrt{8}) + \cosh^{-1} 5 \right) =$$

AP EAPCET 2024 - 19th May Evening Shift

Options:

A. $\sqrt{6} + 4\sqrt{2}$

B. $15 + 8\sqrt{3}$

C. $6\sqrt{6} + 10\sqrt{2}$

D. $8 - 15\sqrt{3}$

Answer: B

Solution:

To solve the expression $\cosh \left(\sinh^{-1}(\sqrt{8}) + \cosh^{-1} 5 \right)$, follow the steps outlined below:

Express the Inverse Hyperbolic Functions as Logarithms:

$$\sinh^{-1}(\sqrt{8}) = \log(\sqrt{8} + \sqrt{8+1})$$

Simplifying:

$$= \log(\sqrt{8} + 3) = \log(3 + \sqrt{8})$$

Similarly, for $\cosh^{-1}(5)$:

$$\cosh^{-1}(5) = \log(5 + \sqrt{5^2 - 1})$$

Simplifying:

$$= \log(5 + \sqrt{24}) = \log(5 + 2\sqrt{6})$$

Combine the Logarithms:

$$\sinh^{-1}(\sqrt{8}) + \cosh^{-1}(5) = \log((3 + \sqrt{8}) \cdot (5 + 2\sqrt{6}))$$

Evaluate the Product Inside the Logarithm:

Calculate:

$$(3 + \sqrt{8})(5 + 2\sqrt{6}) = 3 \cdot 5 + 3 \cdot 2\sqrt{6} + \sqrt{8} \cdot 5 + \sqrt{8} \cdot 2\sqrt{6}$$

$$= 15 + 6\sqrt{6} + 5\sqrt{8} + 2\sqrt{48}$$

Simplifying $\sqrt{8} = 2\sqrt{2}$ and $\sqrt{48} = 4\sqrt{3}$:

$$= 15 + 6\sqrt{6} + 10\sqrt{2} + 8\sqrt{3}$$

Calculate the $\cosh(x)$ Value:

Since $e^x = 15 + 10\sqrt{2} + 6\sqrt{6} + 8\sqrt{3}$, $\cosh(x)$ is given by:

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

Simplify $\cosh(x)$:

$$= \frac{1}{2} \left[(3 + 2\sqrt{2})(5 + 2\sqrt{6}) + (3 - 2\sqrt{2})(5 - 2\sqrt{6}) \right]$$

$$= \frac{1}{2} [30 + 16\sqrt{3}] = 15 + 8\sqrt{3}$$

Thus, the expression evaluates to $15 + 8\sqrt{3}$.

Question41

$$\tan^{-1} 2 + \tan^{-1} 3 =$$

AP EAPCET 2024 - 18th May Morning Shift

Options:

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. $\frac{3\pi}{4}$

D. $\frac{5\pi}{4}$

Answer: C

Solution:

To solve the expression $\tan^{-1} 2 + \tan^{-1} 3$, we can use the formula for the sum of inverse tangents:

$$\tan^{-1} a + \tan^{-1} b = \tan^{-1} \left(\frac{a+b}{1-ab} \right)$$

Applying this formula to our problem, we have:

$$\tan^{-1} 2 + \tan^{-1} 3 = \tan^{-1} \left(\frac{2+3}{1-2 \times 3} \right)$$

Simplify the fraction:

$$= \tan^{-1} \left(\frac{5}{1-6} \right) = \tan^{-1}(-1)$$

We know that $\tan^{-1}(-1) = -\frac{\pi}{4}$. However, we typically express angles in the range 0 to π . Thus:

$$\tan^{-1} 2 + \tan^{-1} 3 = \frac{3\pi}{4}$$

Question42

$\tan^{-1}(-2) - \tan^{-1}(3)$ is equal to

AP EAPCET 2021 - 20th August Evening Shift

Options:

A. $\frac{3\pi}{4}$

B. $\frac{-\pi}{6}$

C. $\frac{\pi}{6}$

D. $\frac{-3\pi}{4}$

Answer: D

Solution:

$$\begin{aligned}\tan^{-1}(-2) - \tan^{-1}(3) &= \tan^{-1} \left[\frac{(-2) - 3}{1 + (-2)(3)} \right] \\ &= \tan^{-1} \left[\frac{-5}{-5} \right] = \tan^{-1} 1\end{aligned}$$

From options, $\tan \frac{3\pi}{4} = -1$

$$\Rightarrow \tan \left(\frac{-3\pi}{4} \right) = 1 \Rightarrow \tan^{-1}(1) = \frac{-3\pi}{4}$$

$$\therefore \tan^{-1}(-2) - \tan^{-1}(3) = \frac{-3\pi}{4}$$

Question43

If $x = \sin (2 \tan^{-1} 2)$, $y = \cos (2 \tan^{-1} 3)$ and $z = \sec (3 \tan^{-1} 4)$, then

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. $x < y < z$

B. $y < z < x$

C. $z < x < y$

D. $z < y < x$

Answer: D

Solution:

Step 1: Analyze $x = \sin(2 \tan^{-1} 2)$

We start with the expression $x = \sin(2 \tan^{-1} 2)$.

Let $\theta = \tan^{-1} 2$, which means that $\tan \theta = 2$. Now, use the double angle identity for sine:

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

We need to find $\sin \theta$ and $\cos \theta$. Since $\tan \theta = \frac{\sin \theta}{\cos \theta} = 2$, we can use the Pythagorean identity to find $\sin \theta$ and $\cos \theta$:

$$\sin^2 \theta + \cos^2 \theta = 1$$

Let $\sin \theta = 2k$ and $\cos \theta = k$. From the identity:

$$(2k)^2 + k^2 = 1 \Rightarrow 4k^2 + k^2 = 1 \Rightarrow 5k^2 = 1 \Rightarrow k = \frac{1}{\sqrt{5}}$$

Thus, $\sin \theta = \frac{2}{\sqrt{5}}$ and $\cos \theta = \frac{1}{\sqrt{5}}$.

Now, using the double angle formula:

$$x = \sin(2\theta) = 2 \left(\frac{2}{\sqrt{5}} \right) \left(\frac{1}{\sqrt{5}} \right) = \frac{4}{5}$$

Step 2: Analyze $y = \cos(2 \tan^{-1} 3)$

Next, for $y = \cos(2 \tan^{-1} 3)$, let $\alpha = \tan^{-1} 3$, so that $\tan \alpha = 3$.

Using the double angle identity for cosine:

$$\cos(2\alpha) = \cos^2 \alpha - \sin^2 \alpha$$

We need to find $\sin \alpha$ and $\cos \alpha$. Since $\tan \alpha = 3$, we use the same method:

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

Let $\sin \alpha = 3k$ and $\cos \alpha = k$, so:

$$(3k)^2 + k^2 = 1 \Rightarrow 9k^2 + k^2 = 1 \Rightarrow 10k^2 = 1 \Rightarrow k = \frac{1}{\sqrt{10}}$$

Thus, $\sin \alpha = \frac{3}{\sqrt{10}}$ and $\cos \alpha = \frac{1}{\sqrt{10}}$.

Now, applying the double angle identity for cosine:

$$y = \cos(2\alpha) = \left(\frac{1}{\sqrt{10}}\right)^2 - \left(\frac{3}{\sqrt{10}}\right)^2 = \frac{1}{10} - \frac{9}{10} = -\frac{8}{10} = -\frac{4}{5}$$

Step 3: Analyze $z = \sec(3 \tan^{-1} 4)$

Lastly, for $z = \sec(3 \tan^{-1} 4)$, let $\beta = \tan^{-1} 4$, so $\tan \beta = 4$.

We need the secant of 3β , and we'll use the triple angle identity for secant:

$$\sec(3\beta) = \frac{1 + 2 \tan^2 \beta}{1 - \tan^2 \beta}$$

For $\tan \beta = 4$, this gives:

$$z = \sec(3\beta) = \frac{1 + 2(4)^2}{1 - (4)^2} = \frac{1 + 2(16)}{1 - 16} = \frac{1 + 32}{-15} = \frac{33}{-15} = -\frac{11}{5}$$

Step 4: Compare the values

We have:

- $x = \frac{4}{5}$
- $y = -\frac{4}{5}$
- $z = -\frac{11}{5}$

Thus, the correct order is:

$$z < y < x$$

Therefore, the correct answer is **D**: $z < y < x$.

Question44

$\frac{d}{dx} \left\{ \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right) \right\}$ is equal to

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. 0

B. $\frac{1}{2}$

C. $\frac{-1}{2}$

D. -1

Answer: D

Solution:

Let $x = \cos 2\theta$

$$\Rightarrow \frac{d\theta}{dx} = -\frac{1}{\sin 2\theta}$$

$$\therefore \frac{d}{dx} \left\{ \sin^{-1} \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right) \right\} = \frac{d}{dx} \sin^2 \theta = \sin 2\theta \frac{d\theta}{dx} = -1$$

Question 45

If $y = \tan^{-1} \left\{ \frac{ax-b}{bx+a} \right\}$, then y' is equal to

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. $\frac{1}{1+x^2} + \frac{a^2}{a^2+b^2}$

B. $\frac{1}{1+x^2}$

C. $\frac{1}{1+\left(\frac{ax-b}{bx+a}\right)^2}$

D. $\frac{bx+a}{1+(ax-b)^2}$

Answer: B

Solution:

$$\begin{aligned} y' &= \left[\frac{1}{1+\left(\frac{ax-b}{bx+a}\right)^2} \right] \frac{(bx+a)a - (ax-b)b}{(bx+a)^2} \\ &= \left[\frac{1}{(a^2+b^2)(x^2+1)} \right] (a^2+b^2) = \frac{1}{x^2+1} \end{aligned}$$

Question46

For how many distinct values of x , the following $\sin \left[2 \cos^{-1} \cot \left(2 \tan^{-1} x \right) \right] = 0$ holds?

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. 8

B. 2

C. 6

D. 4

Answer: B

Solution:

$$\begin{aligned}\sin \left[2 \cos^{-1} \left\{ \cot \left(2 \tan^{-1} x \right) \right\} \right] &= 0 \\ \Rightarrow 2 \cos^{-1} \left\{ \cot \left(2 \tan^{-1} x \right) \right\} &= \sin^{-1}(0) = 0 \\ \Rightarrow \cos^{-1} \left(\cot \left(2 \tan^{-1} x \right) \right) &= 0 \\ \Rightarrow \cot \left[\left(2 \tan^{-1} x \right) \right] &= \cos 0 = 1 \\ \Rightarrow \cot \left[\tan^{-1} \left(\frac{2x}{1-x^2} \right) \right] &= 1 \\ \Rightarrow \cot \left[\cot^{-1} \left(\frac{1-x^2}{2x} \right) \right] &= 1 \\ \Rightarrow \frac{1-x^2}{2x} &= 1 \\ \Rightarrow 1-x^2 &= 2x \\ \Rightarrow x^2 + 2x - 1 &= 0 \\ x &= \frac{-2 \pm \sqrt{4+4}}{2} = \frac{-2 \pm 2\sqrt{2}}{2} \\ x &= -1 \pm \sqrt{2}\end{aligned}$$

Given equation holds for 2 values of x .

Question47

If $\tan^{-1} \left[\frac{1}{1+1 \cdot 2} \right] + \tan^{-1} \left[\frac{1}{1+2 \cdot 3} \right] + \dots + \tan^{-1} \left[\frac{1}{1+n(n+1)} \right] = \tan^{-1}[x]$,
then x is equal to

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $\frac{1}{n+1}$

B. $\frac{n}{n+1}$

C. $\frac{1}{n+2}$

D. $\frac{n}{n+2}$

Answer: D

Solution:

$$\begin{aligned} & \tan^{-1} \left(\frac{1}{1+1 \cdot 2} \right) + \tan^{-1} \left(\frac{1}{1+2 \cdot 3} \right) + \dots \\ & + \tan^{-1} \left(\frac{1}{1+n(n+1)} \right) = \tan^{-1}(x) \\ \Rightarrow & \tan^{-1} \left(\frac{2-1}{1+1 \cdot 2} \right) + \tan^{-1} \left(\frac{3-2}{1+2 \cdot 3} \right) + \dots \\ & + \tan^{-1} \left(\frac{n+1-n}{1+n(n+1)} \right) = \tan^{-1}(x) \\ \Rightarrow & \tan^{-1}(2) - \tan^{-1}(1) + \tan^{-1}(3) - \tan^{-1}(2) + \dots \\ \Rightarrow & \tan^{-1}(n+1) - \tan^{-1}(n) = \tan^{-1}(x) \\ \Rightarrow & \tan^{-1} \left(\frac{n+1-1}{1+(n+1) \cdot 1} \right) = \tan^{-1}(x) = \tan^{-1}(x) \\ \Rightarrow & \tan^{-1} \left(\frac{n}{n+2} \right) = \tan^{-1}(x) \Rightarrow x = \frac{n}{n+2} \end{aligned}$$

Question48

If $y = \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right)$, where $x^2 \leq 1$. Then, find $\frac{dy}{dx}$ is equal to

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $\frac{\pi}{4} + \frac{1}{2} \cos^{-1}(x^2)$

B. $\frac{\pi}{4} - \frac{1}{2} \cos^{-1}(x^2)$

C. $\frac{-x}{\sqrt{1-x^4}}$

D. $\frac{-2x}{\sqrt{1-x^4}}$

Answer: C

Solution:

$$y = \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right)$$

Substituting $x^2 = \cos 2\theta$

$$\Rightarrow \theta = \frac{1}{2} \cos^{-1}(x^2)$$

$$\therefore y = \tan^{-1} \left(\frac{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}} \right)$$

$$= \tan^{-1} \left(\frac{\sqrt{2} \cos \theta + \sqrt{2} \sin \theta}{\sqrt{2} \cos \theta - \sqrt{2} \sin \theta} \right)$$

$$= \tan^{-1} \left(\frac{1 + \tan \theta}{1 - \tan \theta} \right)$$

$$= \tan^{-1} \left(\tan \left(\frac{\pi}{4} + \theta \right) \right) \Rightarrow y = \frac{\pi}{4} + \theta$$

$$\Rightarrow y = \frac{\pi}{4} + \frac{1}{2} \cos^{-1}(x^2)$$

$$\therefore \frac{dy}{dx} = \frac{1}{2} \cdot \left(\frac{-2x}{\sqrt{1-x^4}} \right) = \frac{-x}{\sqrt{1-x^4}}$$

Question49

If $\int \frac{dx}{x(\sqrt{x^4-1})} = \frac{1}{k} \sec^{-1}(x^k)$, then the value of k is equal to

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. 1

B. 2

C. 3

D. 4

Answer: B

Solution:

$$\frac{1}{k} \sec^{-1}(x^k) = \int \frac{dx}{x\sqrt{x^4-1}} = \frac{1}{2} \int \frac{2x dx}{x^2\sqrt{x^4-1}}$$

Let

$$\begin{aligned} \Rightarrow 2x dx &= dt \\ &= \frac{1}{2} \int \frac{dt}{t\sqrt{t^2-1}} = \frac{1}{2} \sec^{-1} t \\ \Rightarrow \frac{1}{k} \sec^{-1}(x^k) &= \frac{1}{2} \sec^{-1}(x^2) \Rightarrow k = 2 \end{aligned}$$
